# Addressing the Closure Problem: State Estimation and Neural Approaches for Partially Observable Dynamical Systems with Structural Model Error

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## 1 Introduction

A fundamental challenge in the modeling of complex dynamical systems is one of closure where the evolution of the resolved variables depends on unresolved states, processes, and scales that are not directly accessible. This mismatch between the “true” system and its analytical or reduced approximation stems from partial observability, missing or inaccurate physics, and parameter uncertainty. The challenge is ubiqutous, appearing in a variety of diverse fields including turbulence modeling, climate science, molecular dynamics, and biological systems.

### 1.1 The Reactivity and Closure Problem

Electromechanical energy harvesters represent a class of ”reactive” multi-domain dynamical systems. The system state $\mathbf{x}(t) \in \mathbb{R}^n$ evolves according to coupled mechanical and electrical laws, driven by external inputs $\mathbf{u}(t)$ (e.g., stochastic vibration sources). A significant challenge in analyzing these systems is the discrepancy between the theoretical dimension of the system and the available data.

While the analytical model typically involves $n \approx 7$ coupled states (representing mechanical modes, electrical potential, and internal flux linkages), the physical realization of the device usually offers limited observability $\mathbf{y}(t) \in \mathbb{R}^m$, where $m \ll n$. Typically, only output voltage or tip displacement is measurable, leaving internal states hidden.

This creates a ”Closure Problem.” Standard engineering approaches assume the analytical model $f_{analytical}$ is structurally correct and only requires parameter estimation ($\theta$). However, experimental evidence suggests the presence of **missing physics**—dynamics not captured by the derived constitutive equations (e.g., unmodeled hysteresis, nonlinear damping, or higher-order coupling). The research objective is to learn a correction term $g_{learn}(\mathbf{x}, \mathbf{u})$ such that:

$$\dot{\mathbf{x}} = f_{analytical}(\mathbf{x}, \mathbf{u}; \theta) + g_{learn}(\mathbf{x}, \mathbf{u}) \tag{1}$$

subject to the constraint that $\mathbf{x}$ is only partially observable via $\mathbf{y} = h(\mathbf{x})$.

## 2 Baseline Methods: State Estimation & Control Theory

### 2.1 Kalman Filtering Variants (EKF/UKF)

State estimation techniques, such as the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF), are the standard for inferring hidden states $\hat{\mathbf{x}}$ from noisy outputs $\mathbf{y}$.

- **Utility:** These methods use a predictor-corrector architecture. The predictor propagates the analytical model, and the corrector updates the state based on the error between predicted and measured output.

- **Limitation in this Context:** EKF assumes that the process noise is zero-mean Gaussian. In this research, the "noise" is actually **structural bias** (the missing physics). When $f_{analytical}$ is incorrect, the Kalman Gain forces the state estimate to compensate for the model error, often leading to biased state estimates rather than identifying the missing dynamic term.

### 2.2 Nonlinear Model Predictive Control (NMPC)

NMPC offers a framework for optimization-based control. While effective for trajectory planning, it typically requires a high-fidelity model to be available *a priori*. In the context of missing physics, NMPC performance degrades rapidly as the prediction horizon increases.

## 3 SciML Frontiers: Investigated Avenues

### 3.1 Universal Differential Equations (UDEs) & Adjoint Optimization

To address the missing physics, we investigate Universal Differential Equations (UDEs), where a Neural Network (NN) approximates the unknown term $g_{learn}$.

$$\dot{\mathbf{x}} = f_{physics}(\mathbf{x}) + \mathrm{NN}(\mathbf{x}; \mathbf{W}) \tag{2}$$

A common misconception is that UDEs require data for all states $\mathbf{x}$. By utilizing **Adjoint Sensitivity Analysis**, we can train the network using only the output $\mathbf{y}$. The error between the predicted output $\hat{\mathbf{y}}$ and measured output $\mathbf{y}_{data}$ is backpropagated through the ODE solver to update the weights $\mathbf{W}$ of the neural network inside the derivative function.

**Algorithm 1** Training UDE with Partial Observability (Adjoint Method)

**Input:** Analytical model $f$, Neural Network $U_\theta$, Output Data $\mathbf{y}_{true}$
**Initialize:** Network weights $\theta$
**while** Loss $>$ tolerance **do**
  Solve forward ODE: $\dot{\mathbf{x}} = f(\mathbf{x}) + U_\theta(\mathbf{x})$ to get trajectory $\mathbf{x}(t)$
  Compute predicted output: $\hat{\mathbf{y}}(t) = h(\mathbf{x}(t))$
  Compute Loss: $\mathcal{L} = ||\hat{\mathbf{y}}(t) - \mathbf{y}_{true}(t)||^2$
  Solve Adjoint ODE backward to get gradients $\nabla_\theta \mathcal{L}$
  Update weights: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
**end while**

### 3.2 Latent Dynamics & Koopman Operator (HAVOK)

Since the true dimension of the "active" physics is unknown, we investigate the Hankel Alternative View of Koopman (HAVOK). By constructing Time-Delay Embeddings from the single output $\mathbf{y}(t)$, we can reconstruct a shadow manifold of the full system. This technique allows for the identification of a linear operator acting on a lifted space, effectively bypassing the need for the analytical model to define the state basis.

# 4 Proposed Novelty: Energy-Constrained & Joint Architectures

## 4.1 Port-Hamiltonian Neural ODEs (PH-NODE)

A critical flaw in standard "black-box" Neural ODEs is the potential violation of conservation laws (e.g., generating free energy). For an energy harvester, energy conservation is the fundamental physical constraint.

We propose investigating a **Port-Hamiltonian framework** where the Neural Network learns the Hamiltonian (Total Energy) function $\mathcal{H}(\mathbf{x})$ and the Damping matrix $R(\mathbf{x})$, rather than the vector field directly. The dynamics are defined as:

$$\dot{\mathbf{x}} = [J(\mathbf{x}) - R(\mathbf{x})]\nabla\mathcal{H}(\mathbf{x}) + G(\mathbf{x})\mathbf{u} \tag{3}$$

where $J(\mathbf{x})$ is the skew-symmetric interconnection structure and $R(\mathbf{x})$ is the positive semi-definite damping structure. This approach guarantees stability and passivity by construction, effectively constraining the "search space" for the missing physics to only physically realizable models.

## 4.2 Multi-Fidelity Transfer Learning

To overcome the lack of hidden state data, a "Sim-to-Real" transfer learning approach is proposed:

1. **Pre-training:** Train the UDE structure on high-fidelity FEM simulation data (where all states $\mathbf{x}$ are observable) to capture the general structure of the nonlinearity.
2. **Transfer:** Freeze the learned architecture of the hidden couplings.
3. **Fine-tuning:** Use the limited experimental output data $\mathbf{y}$ to fine-tune the coefficients of the missing physics term via Adjoint Optimization.

# 5 Future Work and Implementation Plan

The immediate next steps for this research involve:

- Benchmarking the analytical model against experimental data to quantify the baseline residual.
- Implementing a HAVOK analysis to determine the effective dimensionality of the experimental data.
- Developing the Port-Hamiltonian UDE framework within the Julia SciML ecosystem.